

| Method        | Rank1@       |              |              | Rank5@       |              |              |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|
|               | 0.3          | 0.5          | 0.7          | 0.3          | 0.5          | 0.7          |
| MCN           | 39.35        | 21.36        | 6.43         | 68.12        | 53.23        | 29.70        |
| CTRL          | 47.43        | 29.01        | 10.34        | 75.32        | 59.17        | 37.54        |
| TGN           | 43.81        | 27.93        | —            | 54.56        | 44.20        | —            |
| ACRN          | 49.70        | 31.67        | 11.25        | 76.50        | 60.34        | 38.57        |
| CMIN          | <b>63.61</b> | <i>43.40</i> | <i>23.88</i> | <i>80.54</i> | <i>67.95</i> | <i>50.73</i> |
| QSPN          | 52.13        | 33.26        | 13.43        | 77.72        | 62.39        | 40.78        |
| ABLR          | 55.67        | 36.79        | —            | —            | —            | —            |
| TripNet       | 48.42        | 32.19        | 13.93        | —            | —            | —            |
| <b>2D-TAN</b> | Pool         | <i>59.45</i> | <b>44.51</b> | 26.54        | 85.53        | <b>77.13</b> |
|               | Conv         | 58.75        | 44.05        | <b>27.38</b> | <b>85.65</b> | <b>62.26</b> |

Table 2: Performance comparison on ActivityNet Captions.

$IoU@0.5$  and  $0.7$ . It validates that 2D-TAN is able to localize the moment boundary more precisely.

In more details, by comparing 2D-TAN with other related methods, we obtain several observations. First, we compare 2D-TAN with previous sliding window based methods: MCN, CTRL, ACRN, ACL-K and VAL. From the results in Table 1–3, we observe that our 2D-TAN achieves superior results than sliding window methods. The reason is that independently matching the sentence with moment candidates ignores the temporal dependencies, and cannot distinguish the small differences between overlapped moments. Differently, our proposed 2D-TAN models the dependencies between moment candidates by a 2D temporal map, and enables the network to perceive more context information from the adjacent moment candidates. Hence, it gains large improvements compared to sliding window based methods.

Moreover, we compare our approach with RNN-based methods, i.e. TGN and CMIN. Due to the involvement of context information during prediction, the RNN-based approaches perform better than the sliding window approaches, however, inferior to our proposed 2D-TAN method. RNN-based approaches implicitly update the context information through a recurrent memory module, while our 2D-TAN explicitly exploit the context information via a 2D temporal map. It further verifies the effectiveness of our model in high quality moment localization.

Last, we compare our method with graph convolutional network (GCN) based method MAN (?), which achieves the state-of-the-art on Charades-STA. It utilizes a GCN to model the relations between moment pairs. Differently, our 2D-TAN models the temporal dependencies through a 2D convolution network. From Table 1, we can see that 2D-TAN performs better at higher  $IoU@0.7$ , while slightly inferior to MAN at lower  $IoU@0.5$ .

## Ablation Study

In this section, we evaluate the effects of different factors in our proposed 2D-TAN. The experiments are conducted on the ActivityNet Captions dataset, as shown in Table 4.

*Number of Moment Candidates.* The number of moment candidates is a vital factor in moment localization models. We first tune this factor in our 2D-TAN approach, and show

| Method        | Rank1@       |              |              | Rank5@       |              |              |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|
|               | 0.1          | 0.3          | 0.5          | 0.1          | 0.3          | 0.5          |
| MCN           | 14.42        | —            | 5.58         | 37.35        | —            | 10.33        |
| CTRL          | 24.32        | 18.32        | 13.30        | 48.73        | 36.69        | 25.42        |
| MCF           | 25.84        | 18.64        | 12.53        | 52.96        | 37.13        | 24.73        |
| TGN           | 41.87        | 21.77        | 18.9         | 53.40        | 39.06        | <i>31.02</i> |
| ACRN          | 24.22        | 19.52        | 14.62        | 47.42        | 34.97        | 24.88        |
| ROLE          | 20.37        | 15.38        | 9.94         | 45.45        | 31.17        | 20.13        |
| VAL           | 25.74        | 19.76        | 14.74        | 51.87        | 38.55        | 26.52        |
| ACL-K         | 31.64        | 24.17        | <i>20.01</i> | 57.85        | <i>42.15</i> | 30.66        |
| CMIN          | 32.48        | <i>24.64</i> | 18.05        | <i>62.13</i> | 38.46        | 27.02        |
| QSPN          | 25.31        | 20.15        | 15.23        | 53.21        | 36.72        | 25.30        |
| SM-RL         | 26.51        | 20.25        | 15.95        | 50.01        | 38.47        | 27.84        |
| SLTA          | 23.13        | 17.07        | 11.92        | 46.52        | 32.90        | 20.86        |
| ABLR          | <i>34.70</i> | 19.50        | 9.40         | —            | —            | —            |
| SAP           | 31.15        | —            | 18.24        | 53.51        | —            | 28.11        |
| TripNet       | —            | 23.95        | 19.17        | —            | —            | —            |
| <b>2D-TAN</b> | Pool         | <b>47.59</b> | <b>37.29</b> | <b>25.32</b> | 70.31        | <b>57.81</b> |
|               | Conv         | 46.44        | 35.22        | 25.19        | <b>74.43</b> | 56.94        |

Table 3: Performance comparison on TACoS.

its impacts on final performance. Then, we compare different approaches with respect to this factor.

We vary the number of sampled clips  $N$  from 16 to 64 in our 2D-TAN approach. The results are shown in Table 4 (Row 4 – 6). We observe that, increasing  $N$  from 16 to 64 brings improvements (57.79 *v.s* 59.66 *v.s* 60.18 in  $Rank5@0.7$ ). This observation is also consistent with the theoretical upper bound, as listed in Table 4 (Row 1 – 3). Here, the upper bound represents the performance of an ideal model that can provide a correct prediction on all the sampled video clips. The upper bound is smaller than 100% since that the sampling of video clips introduces errors.

Furthermore, we compare the number of moment candidates with the previous state-of-the-art method CMIN. Row 16 in Table 4 shows that CMIN use 1400 ( $N=200$ ) moment candidates. However, our 2D-TAN only uses 136 ( $N=16$ ) candidates, and achieves comparable results to CMIN (Row 4 *v.s* 16 ). Moreover, with larger number of moment candidates ( $N=64$ ) and stacked convolution layers for moment representations, the performance of our method can be further boosted, as listed in Row 14. Noted that the number of moment candidates in Row 14 is 1200, which is still smaller than the ones used in CMIN. This comparison validates that our 2D-TAN gains improvement from the context modeling, rather than the dense sampling of moment candidates.

*Receptive Field Size.* We vary the depth and kernel size of convolution layers in our 2D-TAN to study the impact of receptive field size. The results in terms of different kernel sizes and layer depths are reported in Table 4 Row 7 – 9. We observe that the performance increases significantly as the receptive field enlarges. However, it becomes saturated when it is large enough, as listed in Row 6. Moreover, if the receptive field size is fixed, changing the depth of layers and kernel sizes has limited impacts on final performance, as shown in Row 11–13. This verifies the importance of receptive field size in our 2D-TAN. The large receptive field is able to model temporal dependencies, resulting in perfor-

| Row# | Method      |      | $N$ | 2D-TAN |       | Rank1@ |       |       | Rank5@ |       |       |
|------|-------------|------|-----|--------|-------|--------|-------|-------|--------|-------|-------|
|      |             |      |     | Kernel | Layer | 0.3    | 0.5   | 0.7   | 0.3    | 0.5   | 0.7   |
| 1    | Upper Bound |      | 16  | —      | —     | 97.16  | 93.58 | 89.14 | 97.16  | 93.58 | 89.14 |
| 2    | Upper Bound |      | 32  | —      | —     | 99.10  | 96.88 | 94.38 | 99.10  | 96.88 | 94.38 |
| 3    | Upper Bound |      | 64  | —      | —     | 99.84  | 98.94 | 97.34 | 99.84  | 98.94 | 97.34 |
| 4    | 2D-TAN      | Enum | 16  | 9      | 4     | 58.82  | 42.45 | 23.93 | 85.07  | 75.99 | 57.79 |
| 5    |             | Enum | 32  | 9      | 4     | 58.26  | 43.18 | 25.47 | 84.82  | 75.45 | 59.66 |
| 6    |             | Enum | 64  | 9      | 4     | 58.15  | 42.80 | 25.76 | 84.53  | 75.39 | 60.18 |
| 7    |             | Enum | 64  | 1      | 1     | 45.90  | 26.20 | 14.27 | 70.72  | 56.14 | 37.13 |
| 8    |             | Enum | 64  | 5      | 1     | 54.78  | 35.27 | 18.81 | 81.80  | 69.76 | 50.68 |
| 9    |             | Enum | 64  | 5      | 4     | 58.20  | 40.45 | 23.25 | 83.76  | 73.97 | 57.46 |
| 10   |             | Enum | 64  | 9      | 4     | 58.15  | 42.80 | 25.76 | 84.53  | 75.39 | 60.18 |
| 11   |             | Pool | 64  | 9      | 4     | 59.45  | 44.51 | 26.54 | 85.53  | 77.13 | 61.96 |
| 12   |             | Pool | 64  | 5      | 8     | 57.86  | 41.68 | 25.13 | 85.26  | 75.74 | 58.90 |
| 13   |             | Pool | 64  | 17     | 2     | 58.19  | 43.09 | 26.09 | 84.22  | 75.16 | 60.02 |
| 14   |             | Conv | 64  | 9      | 4     | 58.75  | 44.05 | 27.38 | 85.65  | 76.65 | 62.26 |
| 15   | CTRL        |      | —   | —      | —     | 47.43  | 29.01 | 10.34 | 75.32  | 59.17 | 37.54 |
| 16   | CMIN        |      | 200 | —      | —     | 63.61  | 43.40 | 23.88 | 80.54  | 67.95 | 50.73 |

Table 4: Ablation Study.  $N$  is the number of sampled clips. Row 1 – 3 show the upper bound of an ideal model under different  $N$ . Row 4 – 6 demonstrate how our model perform under different  $N$ . Row 6 – 13 compare the performance under different kernel and layer settings. Row 14 show the performance using moment features extracted by stacked convolution. Row 15 – 16 are two previous methods for comparison.

mance improvements. If we set the kernel size to 1 (Row 7), the 2D-TAN model is equivalent to treat each moment independently. In this case, it achieves similar performance with CTRL method (Row 15), which also treats each moment individually. This phenomenon further proves our hypothesis that modeling the moment candidates as a whole enables the network to distinguish similar moments. *Sparse Sampling* v.s. *Enumeration*. We further compare the effectiveness of our sparse sampling strategy with the dense enumeration for moment candidate selection. The results are reported in Table 4 (Row 10-11). It is observed that these two strategies achieve similar performance. The underlying reason is that the designed sparse sampling removes nearly 50% redundant moment candidates. Thus, it reduces the computation cost without performance decrease. *Stacked Convolution* v.s. *Max-Pooling*. Stacked convolution and pooling have been applied for extracting moment features in previous works (?; ?). We compare their performance on three datasets, as shown in Table 1-3 (2D-TAN: Pool v.s. Conv). It is observed that stacked convolution (Conv) performs better than max-pooling (Pool) on ActivityNet Captions, while comparable on Charades-STA and TACoS. We recommend to adopt the max-pooling operation, since it is fast in calculation, while does not contain any parameters.

## Conclusion

In this paper, we study the problem of moment localization with natural language, and propose a novel 2D Temporal Adjacent Networks(2D-TAN) method. The core idea is to retrieve a moment on a two-dimensional temporal map, which considers adjacent moment candidates as the temporal context. 2D-TAN is capable of encoding adjacent temporal relation, while learning discriminative feature for matching video moments with referring expressions. Our model is simple in design and achieves competitive performance

in comparison with the state-of-the-art methods on three benchmark datasets. In the future, we would like to extend our model to other temporal localization tasks, such as temporal action localization, video re-localization, etc.

## Acknowledgement

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